

Causal Diagrams and Identifying Causal Effects

Directed Acyclic Graphs (DAGs)

ECON3500: Econometrics and Applications

2026-01-01

Learning Objectives

By the end of this lecture, you will be able to:

1. Draw and interpret directed acyclic graphs (DAGs)
2. Identify causal paths, backdoor paths, and colliders
3. Determine which variables to control for to identify a causal effect
4. Explain why controlling for the wrong variables can bias your estimates
5. Apply DAG logic to real research questions

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Reading

Huntington-Klein, *The Effect*: **Chapters 6, 7, and 8** (available free at theeffectbook.net)

What Is Causality?

The Data Generating Process

Every dataset we observe is produced by a data generating process (DGP) — the real-world laws, behaviors, and institutions that determine the values we see.

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Our challenge as econometricians:

- We observe the *data*, not the DGP
- Multiple DGPs could produce the same data patterns

- We need a framework for reasoning about *which variation* in our data answers our research question

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! Identification

Identification is the process of figuring out what part of the variation in your data answers your research question.

Association vs. Causation

i Association

Two variables move together (correlation). X and Y are related statistically.

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i Causation

Changing X **causes** Y to change. Formally: if we could *intervene* to change X , the distribution of Y would change as a result.

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- **Association:** Useful for prediction, not necessarily for policy
- **Causation:** Required for policy impact, program evaluation, treatment effects

The Framework: Causal Diagrams (DAGs)

A **directed acyclic graph (DAG)** is a visual representation of a data generating process. Every word in the name matters:

- **Directed:** Arrows point in one direction — from cause to effect
- **Acyclic:** No cycles — you can never follow arrows and end up back where you started
- **Graph:** A diagram with nodes (variables) and edges (arrows)

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The acyclic part is a real constraint: DAGs assume causality flows forward. If X causes Y and Y simultaneously causes X , a DAG can't represent that. (We'll return to this when we discuss simultaneity.)

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Key insight: Once we draw the DAG, we can **mechanically determine** which variables to control for to identify a causal effect.

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Important assumption

Every variable and arrow that is *not* on the diagram is an assumption we're making. Drawing a DAG forces you to be explicit about your identifying assumptions.

Drawing and Reading DAGs

DAG Basics: Nodes and Arrows

Node: A variable in the system.

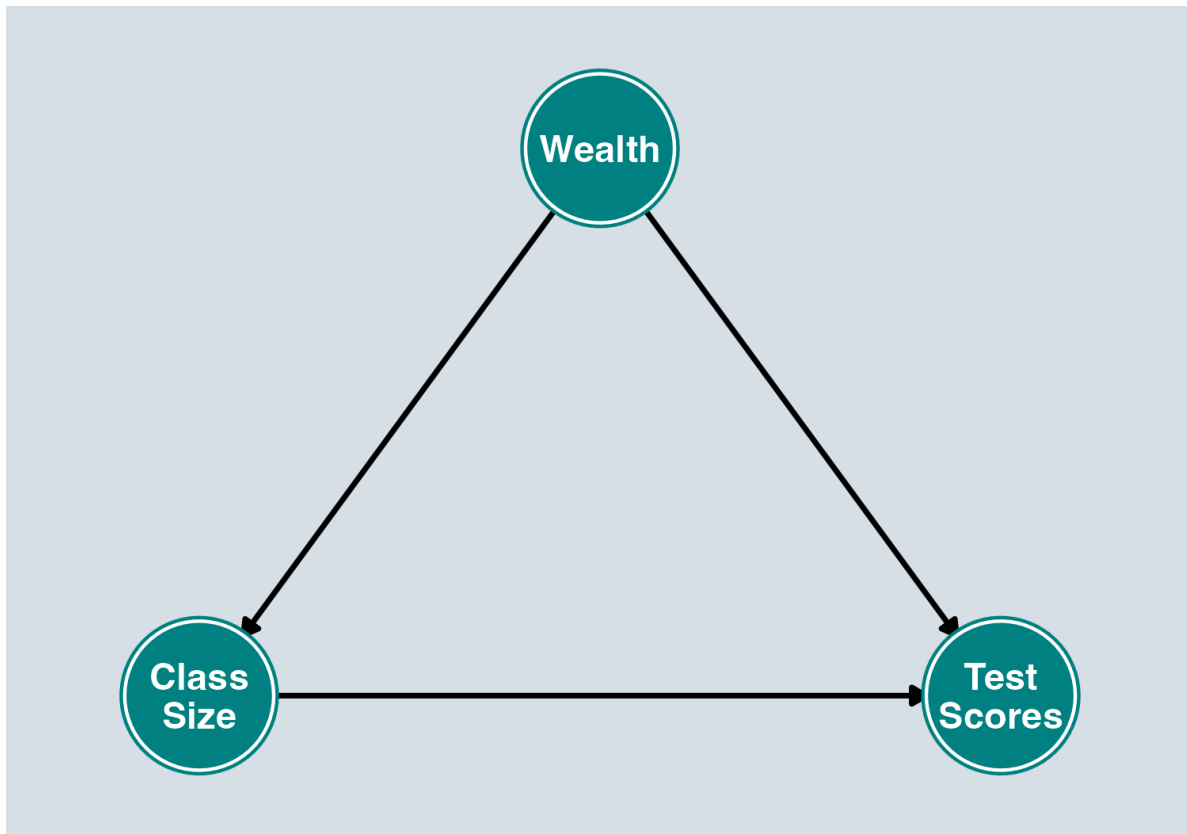
Each circle represents a variable — either observed or unobserved.

Arrow: $X \rightarrow Y$ means “ X causes Y .”

- Direction matters (from cause to effect)
- X is a parent (ancestor) of Y
- Y is a child (descendant) of X



Example: Class Size and Student Achievement



Variables:

- X = Class Size, Y = Test Scores, Z = Wealth

Causal arrows:

- Wealth $\rightarrow$ Class Size (wealthy areas have smaller classes)
- Wealth $\rightarrow$ Test Scores (wealthier students score higher)
- Class Size $\rightarrow$ Test Scores (smaller classes improve learning)

How to Build a DAG

Step-by-Step DAG Construction

Building a DAG is a structured process (HK, *The Effect*, Ch. 6):

Step 1: List all relevant variables. What might cause your treatment? What might cause your outcome? What affects both? Cast a wide net.

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Step 2: For each pair of variables, ask “does one cause the other?” If yes, draw an arrow from cause to effect. If you draw no arrow, you’re assuming no direct causal link between those variables.

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Step 3: Simplify. Remove any variable that is not on *any* path between your treatment and outcome. If it doesn’t connect X and Y (directly or indirectly), it doesn’t affect your identification — so it can be dropped.

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Step 4: Determine what to control for. Apply the backdoor criterion to find the minimal set of controls that closes all backdoor paths. (We’ll get to this soon!)

Building DAGs: Handling Correlated Variables

💡 What if two variables are correlated but neither causes the other?

Sometimes you have two variables — say, Income and SES — that are clearly correlated, but you don’t think one directly causes the other.

Solution: Introduce a shared unobserved cause U that drives both.

$$U \rightarrow \text{Income} \quad \text{and} \quad U \rightarrow \text{SES}$$

This is more honest than arbitrarily picking a direction for the arrow. You’re saying: “something generates both of these, but I don’t need to name it.” In a DAG, draw U as a gray (unobserved) node with arrows pointing to both variables.

Causal (Front Door) Paths

$$X \rightarrow Y \text{ (or } X \rightarrow M \rightarrow Y)$$

Direct or indirect effect of X on Y , with all arrows pointing *away from* X .

These are good paths — they represent what we *want* to estimate.

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Example: Class Size $\rightarrow$ Test Scores

Backdoor Paths

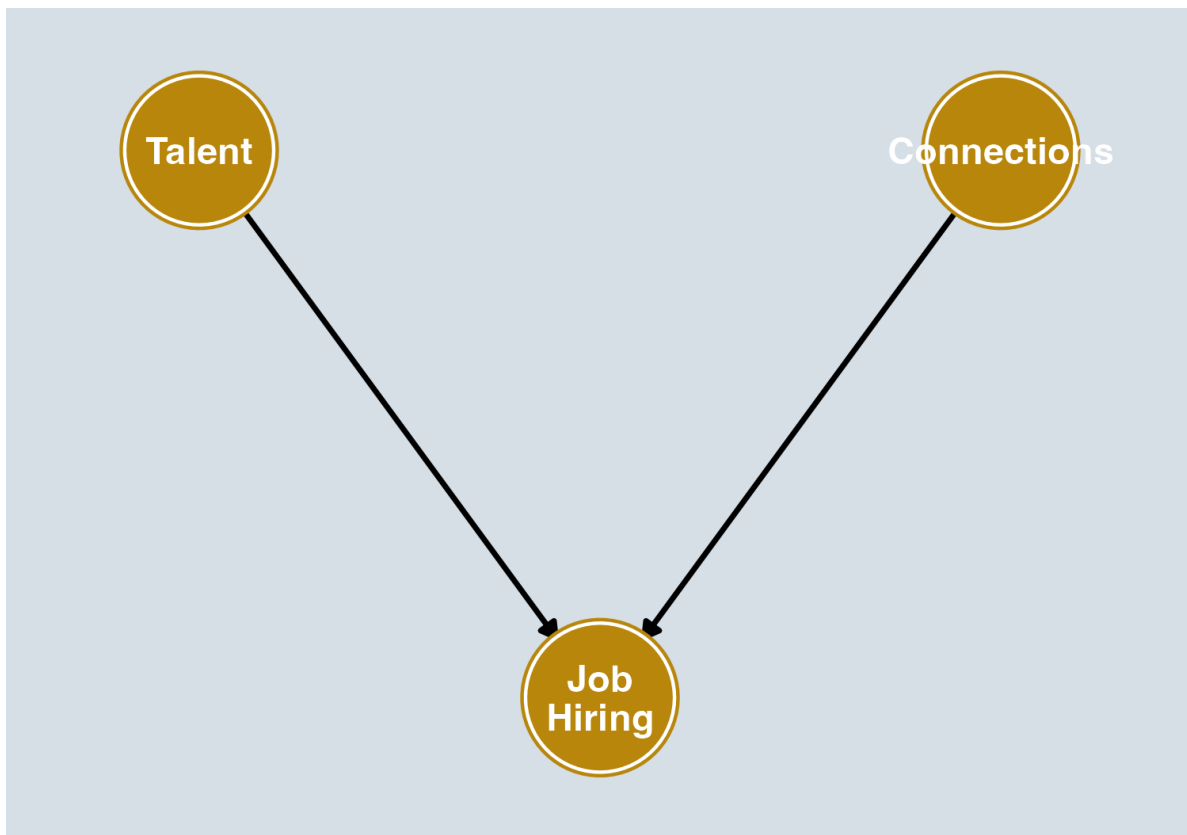
$$X \leftarrow Z \rightarrow Y$$

X and Y are connected through a common cause Z . These are bad paths — they create **confounding**.

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Example: Class Size $\leftarrow$ Wealth $\rightarrow$ Test Scores

Collider Paths



$$X \rightarrow Z \leftarrow Y$$

X and Y both cause Z . These paths are closed by default — no confounding flows through them *unless* we control for Z .

- Talent and Connections both cause Job Hiring

- The path $\text{Talent} \rightarrow \text{Job Hiring} \leftarrow \text{Connections}$ is closed by default
- If we condition on who was hired (control for the collider), the path **opens**
- Among the hired: low Talent becomes correlated with high Connections

What Does “Controlling” Mean?

When we say “control for Z ” in a regression, we are comparing units that have the same value of Z . This removes the variation in X and Y that is driven by Z .

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In DAG terms: Controlling for a variable blocks all paths that pass through it. This can be good or bad:

Path Type	Default Status	What controlling does
Backdoor ($X \leftarrow Z \rightarrow Y$)	Open	Closes it $\rightarrow$ removes bias
Collider ($X \rightarrow Z \leftarrow Y$)	Closed	Opens it $\rightarrow$ creates bias
Mediator ($X \rightarrow Z \rightarrow Y$)	Open	Closes it $\rightarrow$ blocks the causal effect

...

Our goal: Close all bad (backdoor) paths while keeping good (causal) paths open — and not accidentally opening closed ones.

Confounding, OVB, and the Backdoor Criterion

Confounding = OVB = Open Backdoor Path

! Confounding

When X and Y share a common cause Z , they are **confounded**. A regression of Y on X will pick up both the causal effect AND the confounding association.

$$\text{Association} = \underbrace{\text{Causal Effect}}_{\text{Class Size} \rightarrow \text{Scores}} + \underbrace{\text{Confounding Bias}}_{\text{via Wealth}}$$

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Remember omitted variable bias from Chapter 6? **Every case of OVB is an open backdoor path.** The omitted variable creates a backdoor path that stays open, and the bias flows through it into $\hat{\beta}_1$.

Knowledge Check: Direction of Bias

💡 Think About It

If we regress Test Scores on Class Size *without* controlling for Wealth, will the class size coefficient be biased up or down?

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Answer: The coefficient on class size is biased *away from zero* (overstates the negative effect).

- Wealthy areas → smaller classes AND higher scores
- This creates a *negative* confounding association between class size and scores
- The negative confounding reinforces the true negative causal effect, making OLS more negative than the true β_1

The Backdoor Criterion

To identify the causal effect of X on Y :

! Backdoor Criterion

Control for a set of variables $\mathbf{Z}$ such that:

1. No variable in $\mathbf{Z}$ is a descendant of X
2. Controlling for $\mathbf{Z}$ closes all backdoor paths from X to Y

...

In plain language:

- **Do** control for common causes (confounders) — they create open backdoor paths
- **Don't** control for variables that X affects — that would block part of the causal effect
- **Don't** control for colliders — that would *open* a path that was safely closed

Applying the Criterion: List All Paths

A systematic approach to finding your controls:

1. **List every path** connecting X and Y (follow arrows in *any* direction)
2. **Classify each path:** Is it a causal (front-door) path or a backdoor path?
3. **Check each path's status:** Is it currently open or closed?
4. **Choose controls** to close all open backdoor paths — without opening any closed paths

. . .

Example: Class Size and Test Scores

Path	Type	Default	Action
Class Size $\rightarrow$ Test Scores	Causal	Open	Leave open
Class Size $\leftarrow$ Wealth $\rightarrow$ Test Scores	Backdoor	Open	Control for Wealth

After controlling for Wealth: the only remaining open path is causal. The effect is identified.

What NOT to Control For: Mediators



Bad Control: Mediator

If $X \rightarrow M \rightarrow Y$, then M is a mediator. Controlling for M blocks part of the causal effect of X on Y .

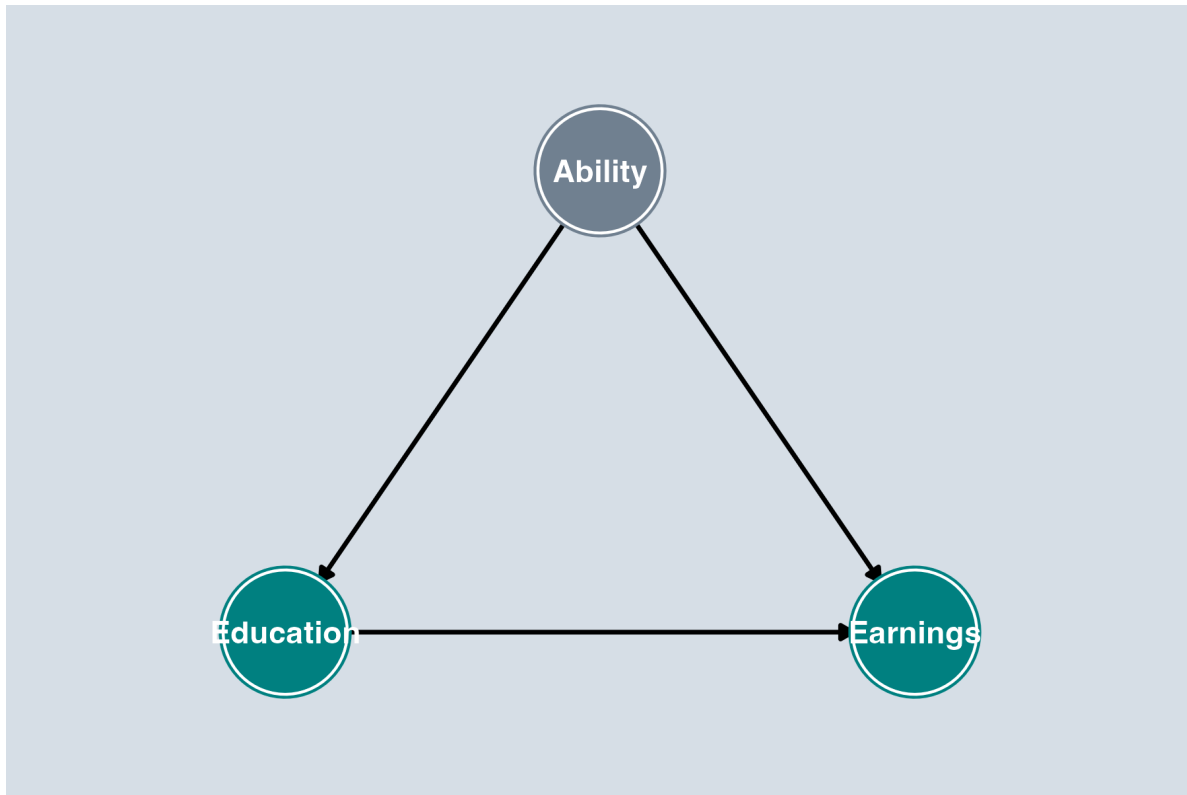
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Example: Does education affect earnings?

- Education $\rightarrow$ Job Type $\rightarrow$ Earnings
- If we control for Job Type, we block the indirect effect of education that works *through* job placement
- We'd only estimate the effect of education holding job type fixed — not the total causal effect

Applications: Identifying Causal Effects

Does Education Cause Earnings?



Gray node = unobserved variable

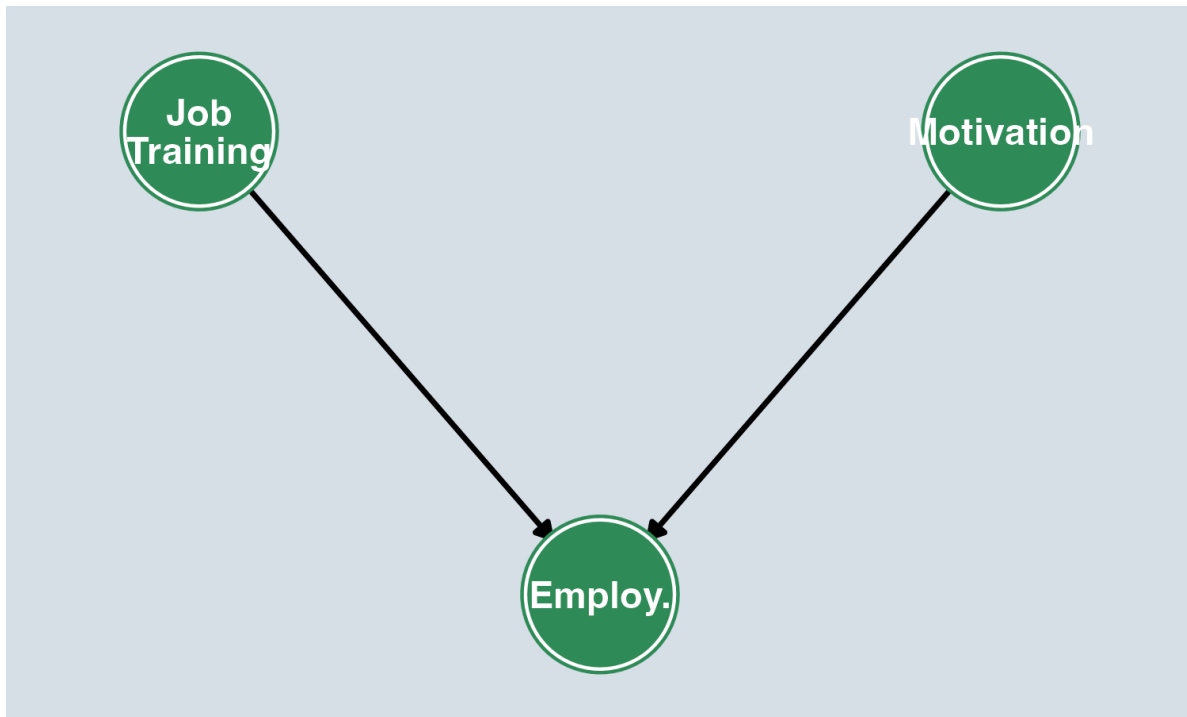
1. **Causal path?** Yes: $\text{Education} \rightarrow \text{Earnings}$
2. **Backdoor paths?** Yes: $\text{Education} \leftarrow \text{Ability} \rightarrow \text{Earnings}$
3. **What to control for?** Ability

Does Education Cause Earnings? (cont.)

Challenge: Ability is unobserved! We can't directly control for it.

This is why economists turn to **identification strategies** — instruments, experiments, panel data — to close backdoor paths when confounders are unobservable.

Does Job Training Improve Employment?



The problem without randomization:

- Motivated workers are more likely to sign up for training
- Motivated workers are also more likely to find jobs
- Backdoor path: Training $\leftarrow$ Motivation $\rightarrow$ Employment

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With an RCT:

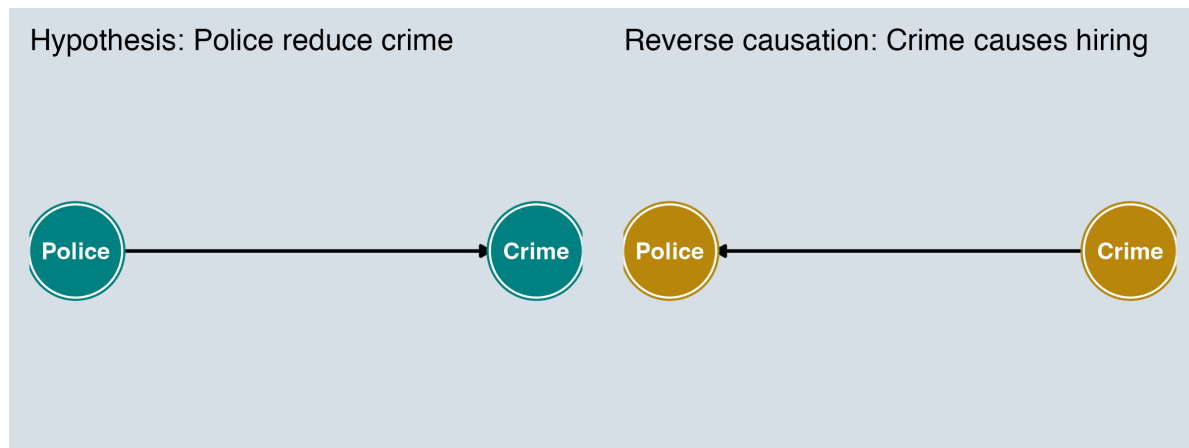
- Randomly assign workers to training
- Now Motivation does not cause Training — that arrow is severed
- All backdoor paths are closed *by design*

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! Important

This is why randomized controlled trials (RCTs) are the gold standard for causal inference — they eliminate confounding mechanically.

Reverse Causation



⚠ Reverse Causation

Sometimes the arrow between X and Y runs in the **opposite direction** from what we assume.

- We observe a positive correlation between police and crime
- **Our hypothesis:** More police $\rightarrow$ less crime
- **The problem:** Cities with more crime hire more police

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In DAG terms: we've drawn the wrong DAG. The true DGP has the arrow reversed — or both arrows exist.

DAGs and Simultaneity

Remember: the “A” in DAG stands for acyclic. DAGs **cannot represent simultaneity** — when X causes Y and Y simultaneously causes X .

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Two approaches when you suspect reverse causation or simultaneity:

💡 Approach 1: Use time subscripts

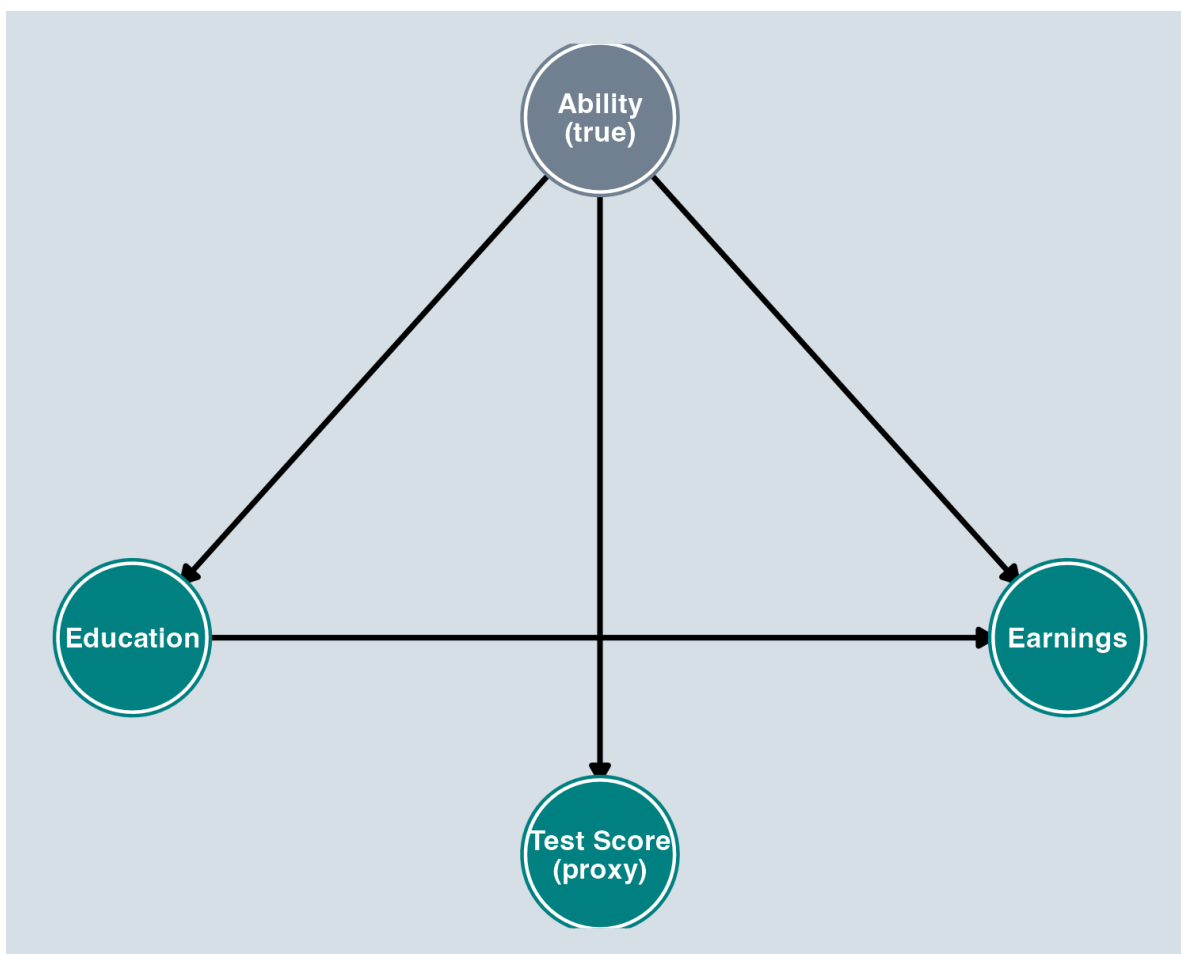
Separate variables by time period: $X_t \rightarrow Y_{t+1}$ and $Y_t \rightarrow X_{t+1}$. This converts a cycle into an acyclic graph — causality flows forward through time. Works well when the feedback loop operates with a lag (e.g., crime this year $\rightarrow$ police budget next year).

. . .

💡 Approach 2: Use instrumental variables

When the relationship is truly simultaneous (e.g., price and quantity in a market), the DAG framework alone can't solve it. This is where instrumental variables and 2SLS become the right tools — you need an external source of variation that shifts one variable without directly affecting the other. We'll cover this in detail later in the course.

Measurement Error



Gray = unobserved

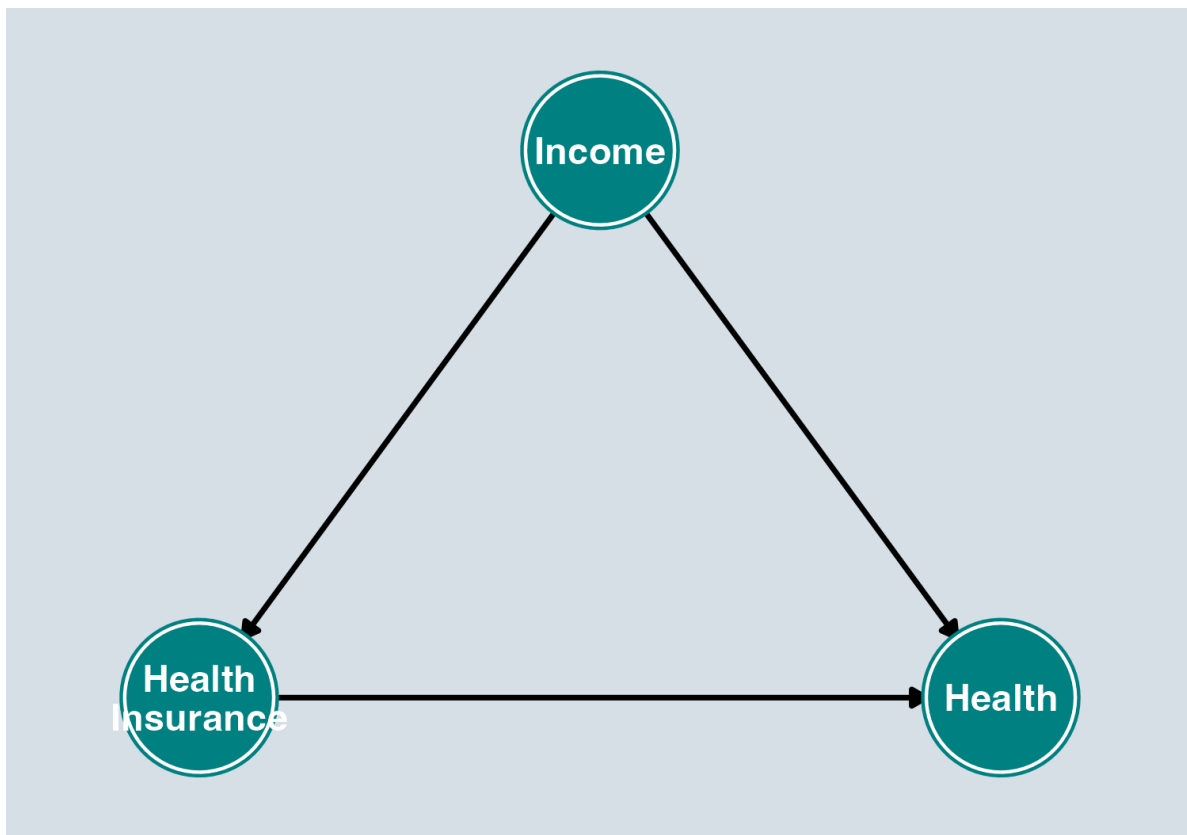
⚠ Measurement Error

When we can't perfectly measure a variable in our DAG, we introduce measurement error.

- We want to control for Ability to close a backdoor path
- We can't measure Ability directly, so we use a proxy (Test Score)
- But Test Score $\rightarrow$ Ability (noisy measurement)
- Controlling for the proxy only *partially* closes the backdoor path
- Result: our estimate is still biased — attenuation bias

Knowledge Checks and Practice

Knowledge Check 1: Health Insurance



Does Income confound the effect of Health Insurance on Health?

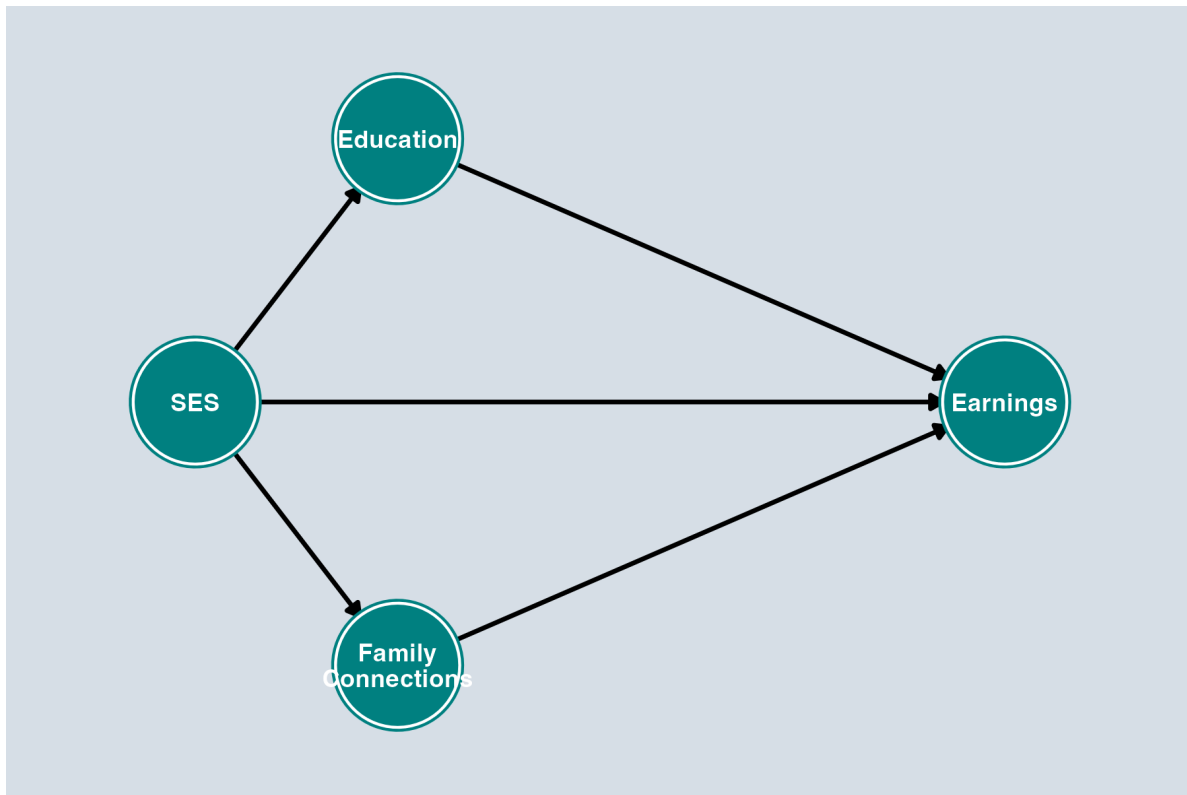
To estimate the effect of Health Insurance on Health, should we control for Income?

Knowledge Check 1: Answer

💡 Answer

- There is a backdoor path: $\text{Health Insurance} \leftarrow \text{Income} \rightarrow \text{Health}$
- Income is a common cause of both Health Insurance and Health
- So yes, we **should** control for Income — it's a confounder

Knowledge Check 2: SES and Earnings



Questions:

1. What are the backdoor paths from SES to Earnings?
2. What minimal set of variables should we control for?

Knowledge Check 2: Answer

Answer

1. **Backdoor path:** $SES \leftarrow (\text{shared causes}) \rightarrow \text{Earnings via Family Connections}$
2. **Controls:** Family Connections (to close the backdoor)
 - Do NOT control for Education — it is a *descendant* of SES (mediator)

Case Study: Does Wine Improve Heart Health?

THE LANCET, MAY 12, 1979	1017																		
Public Health FACTORS ASSOCIATED WITH CARDIAC MORTALITY IN DEVELOPED COUNTRIES WITH PARTICULAR REFERENCE TO THE CONSUMPTION OF WINE A. S. ST. LEGER A. L. COCHRANE* F. MOORE Medical Research Council Epidemiology Unit, Cardiff CF2 3AS Summary Deaths from ischaemic heart-disease in 18 developed countries are not strongly associated with health-service factors such as doctor and nurse density. There is a negative association with gross national product per capita and a positive but inconsistent association with saturated and monounsaturated fat intake. The principal finding is a strong and specific negative association between ischaemic heart-disease deaths and alcohol consumption. This is shown to be wholly attributable to wine consumption.	TABLE 1—COUNTRIES EXAMINED IN THIS STUDY <table><tr><td>Australia</td><td>Republic of Ireland</td></tr><tr><td>Austria</td><td>Italy</td></tr><tr><td>Belgium</td><td>Netherlands</td></tr><tr><td>Canada</td><td>New Zealand</td></tr><tr><td>Denmark</td><td>Norway</td></tr><tr><td>England and Wales</td><td>Scotland</td></tr><tr><td>Finland</td><td>Sweden</td></tr><tr><td>France</td><td>Switzerland</td></tr><tr><td>German Federal Republic</td><td>United States of America</td></tr></table> METHODS The 18 countries used in this study are listed in table 1. The criteria for their selection are stated in our previous work.¹ Our mortality data consists of death-rates for males and females separately in the 55–64 age-group from hypertensive disease, ischaemic heart-disease (I.H.D.), cerebro-vascular disease, and bronchitis plus emphysema and asthma; and in the 25–34 age group death-rates from road accidents. These death-rates were obtained from the World Health Statistics Annual for 1970 (codes A82, A83, A85, A93, and AE138, respectively). The latter two causes of death were included in the study for comparison with heart-disease. It has been suggested² that France underreports I.H.D. deaths and we made allowance for this by analysing our data with France included and excluded. Our health-service, economic, and demographic variables have been described elsewhere,¹ as have cigarette consumption	Australia	Republic of Ireland	Austria	Italy	Belgium	Netherlands	Canada	New Zealand	Denmark	Norway	England and Wales	Scotland	Finland	Sweden	France	Switzerland	German Federal Republic	United States of America
Australia	Republic of Ireland																		
Austria	Italy																		
Belgium	Netherlands																		
Canada	New Zealand																		
Denmark	Norway																		
England and Wales	Scotland																		
Finland	Sweden																		
France	Switzerland																		
German Federal Republic	United States of America																		

Figure 1: St Leger, Cochrane, & Moore (1979). *The Lancet*.

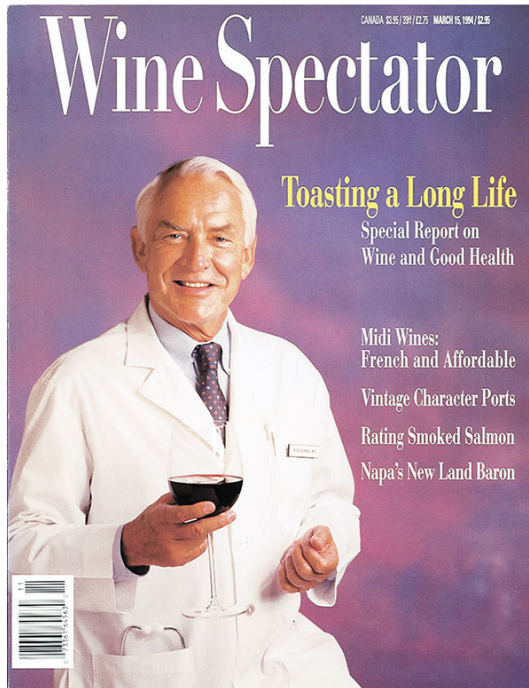
In 1979, researchers found a strong negative association between wine consumption and heart disease across 18 developed countries.

The French Paradox Goes Mainstream



Figure 2: 60 Minutes, November 17, 1991

Wine Steals America's Heart



The March 15, 1994, issue

The mid-1980s found America gripped by a fitness craze, with a Jane Fonda cassette in every VCR and, thanks to the revelation that wine could offer health benefits, a bottle of red wine on more and more dinner tables across the country. Thanks in large part to a *60 Minutes* segment in 1991 that introduced the Western world to the "French paradox," red wine sales in the United States jumped nearly 40 percent in 1992, and Americans had the money to spend. The 1990s brought a booming economy and an increased demand for luxury products. And Napa Valley was primed to deliver: Cult Cabernets

became American wine's success (and excess) story of the decade, epitomized by the debut of Screaming Eagle with the 1992 vintage.

Figure 3: Wine Spectator, March 15, 1994

Red wine sales in the US jumped nearly 40% in 1992. The idea that wine was good for your heart became conventional wisdom.

But Wait...

But [some researchers](#) had been pointing out problems with these kinds of studies since the 1980s, and questioning if the alcohol was responsible for the benefits they saw.

Perhaps moderate drinkers were healthier than non-drinkers, they said, because they were [more likely to be educated](#), wealthy and physically active, and more likely to have health insurance and eat more vegetables. Or maybe, these researchers added, it was because many of the “non-drinkers” in the studies were actually ex-drinkers who had quit because they had developed health issues.

Moderate wine drinkers tend to be **more educated, wealthier, more physically active**, and more likely to have health insurance.

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And many “non-drinkers” were actually **ex-drinkers who had quit because of health problems**.

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In DAG terms: What kinds of bias are these?

Better Methods, Different Answers



Original Investigation | Cardiology

Association of Habitual Alcohol Intake With Risk of Cardiovascular Disease

Kiran J. Biddinger; Connor A. Emdin, MD, DPhil; Mary E. Haas, PhD; Minxian Wang, PhD; George Hindy, MD; Patrick T. Ellinor, MD, PhD; Sekar Kathiresan, MD; Amit V. Khera, MD, MSc; Krishna G. Aragam, MD, MS

Abstract

IMPORTANCE Observational studies have consistently proposed cardiovascular benefits associated with light alcohol consumption, while recent genetic analyses (ie, mendelian randomization studies) have suggested a possible causal link between alcohol intake and increased risk of cardiovascular disease. However, traditional approaches to genetic epidemiology assume a linear association and thus have not fully evaluated dose-response estimates of risk across different levels of alcohol intake.

OBJECTIVES To assess the association of habitual alcohol intake with cardiovascular disease risk and to evaluate the direction and relative magnitude of cardiovascular risk associated with different amounts of alcohol consumption.

DESIGN, SETTING, AND PARTICIPANTS This cohort study used the UK Biobank (2006-2010, follow-up until 2016) to examine confounding in epidemiologic associations between alcohol intake and cardiovascular diseases. Using both traditional (ie, linear) and nonlinear mendelian randomization, potential associations between alcohol consumption and cardiovascular diseases (eg, hypertension and coronary artery disease) as well as corresponding association shapes were assessed. Data analysis was conducted from July 2019 to January 2022.

Key Points

Question What is the risk of cardiovascular disease associated with different amounts of habitual alcohol consumption?

Findings In this cohort study of 371 463 individuals, genetic evidence supported a nonlinear, consistently risk-increasing association between all amounts of alcohol consumption and both hypertension and coronary artery disease, with modest increases in risk with light alcohol intake and exponentially greater risk increases at higher levels of consumption.

Meaning In this study, alcohol consumption at all levels was associated with increased risk of cardiovascular

Figure 4: Biddinger et al. (2022)

Using Mendelian randomization — genetic variants as instruments — alcohol at *all* levels was associated with **increased** cardiovascular risk.



Original Investigation | Public Health

Alcohol Consumption Patterns and Mortality Among Older Adults With Health-Related or Socioeconomic Risk Factors

Rosario Ortola, MD, PhD; Mercedes Sotos-Prieto, PhD; Esther García-Esquinas, PhD; Iñaki Galán, PhD; Fernando Rodríguez-Artalejo, PhD

Abstract

IMPORTANCE Alcohol consumption is a leading cause of morbidity and mortality that may be more important in older adults with socioeconomic or health-related risk factors.

OBJECTIVE To examine the association of alcohol consumption patterns with 12-year mortality and its modification by health-related or socioeconomic risk factors.

DESIGN, SETTING, AND PARTICIPANTS This prospective cohort study used data from the UK Biobank, a population-based cohort. Participants were current drinkers aged 60 years or older. Data were analyzed from September 2023 to May 2024.

EXPOSURE According to their mean alcohol intake in grams per day, participants' drinking patterns were classified as occasional: ≤ 2.86 g/d, low risk (men: >2.86 - 20.00 g/d; women: >2.86 - 10.00 g/d), moderate risk (men: >20.00 - 40.00 g/d; women: >10.00 - 20.00 g/d) and high risk (men: >40.00 g/d; women: >20.00 g/d).

Key Points

Question Do health-related or socioeconomic risk factors modify the associations of alcohol consumption patterns with mortality among older drinkers?

Findings This cohort study in 135 103 older drinkers found that even low-risk drinking was associated with higher mortality among older adults with health-related or socioeconomic risk factors. Wine preference and drinking only with meals were associated with attenuating the excess mortality associated with alcohol consumption.

Figure 5: Ortola et al. (2024)

Even among “low-risk” drinkers, alcohol was associated with higher mortality when accounting for health and SES risk factors.

Your Turn: Wine and Health Worksheet

Now apply DAG thinking to this question using the **worksheet handout**:

1. List all relevant variables
2. Draw arrows between them
3. Simplify your DAG
4. List all paths from Wine to Heart Health
5. Determine what to control for

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Tip from *The Effect*

Start with your treatment and outcome. Then ask: “What are all the things that cause my treatment? What are all the things that cause my outcome? Do any of those overlap?”

Those overlapping causes are your confounders.

Summary

Key Takeaways

1. **The DGP is what we're trying to understand.** A DAG is our best model of that process.
2. **Association Causation.** DAGs help us think systematically about *why*.
3. **OVB = an open backdoor path.** If a confounder is omitted, the path stays open and your estimate is biased.
4. **The Backdoor Criterion is mechanical.** Once you draw the DAG, you can determine what to control for.
5. **Collider bias is real.** Controlling for the wrong variable can *introduce* bias by opening a closed path.
6. **Randomization is powerful.** Random assignment closes backdoor paths by design.

What's Next?

These ideas connect directly to **Chapter 9: Assessing Regression Validity**:

- Omitted variable bias = an open backdoor path you haven't closed
- Measurement error = your DAG nodes don't match what you actually measured
- Simultaneity / reverse causation = the arrows in your DAG might be wrong

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The DAG framework gives you a visual language for diagnosing every threat to internal validity.

For the Curious: d-Separation

The backdoor criterion is actually a special case of a more general concept called d-separation.

- Two variables are **d-separated** by **Z** if every path between them is closed after conditioning on **Z**
- If d-separated, the variables are conditionally independent given **Z**

You don't need to know d-separation for this course — the backdoor criterion is the tool you'll use. But if you want to go deeper, see *The Effect* Chapter 8 or Pearl (2009).